

Exploring a database

David Moreno

Explore the RStudio screen

You can design your own RStudio screen [Pane layout](#)

- Source: write anything (comments and codes). Click on Run to execute.
- Console: see the results.
- Environment: explore the variables and databases
- Plots: show graphs

A dataframe

Import a database in R using the readxl package and get an overview

```
library(readxl)
myeloma <- read_excel("/Users/dfmoreno/Dropbox/Lectures/myeloma.xlsx")
str(myeloma)
```

Examine columns or variables. Use the tab key to quickly see columns

```
myeloma$Patient # myeloma[1] or myeloma["Patient"]
myeloma$molecular_group # myeloma[2] or myeloma["molecular_group"]
```

Examine columns and rows

```
# dataframe[row,column]
myeloma[3,2] # myeloma[3,"molecular_group"]

# Entire rows
myeloma[3,] # entire 3rd row

# If rownames have actual names, do the same as a column
# I do not recommend it
myeloma["GSM51000",]
```

Categorical data

```
class(myeloma$chr1q21_status) # Which type of variable is this?
```

```
[1] "character"
```

```
table(myeloma$chr1q21_status) # Categories
```

```

2 copies  3 copies 4+ copies
      93      55      38
```

Numerical data

```
class(myeloma$CCND1)
```

```
[1] "numeric"
```

```
mean(myeloma$CCND1)
```

```
[1] 4493.613
```

```
median(myeloma$CCND1)
```

```
[1] 693.05
```

Introduction to the tidyverse

Tidyverse is a mix of key packages for data analysis

ggplot: Plotting

lubridate: Dates

dplyr: Data manipulation

tidyr: Reshape databases

Tidyverse works with pipes (`%>%`) that allow you to concatenate analyses. There are multiple functions that can make your life easier: [Examples](#)

```
# Install tidyverse
# install.packages("tidyverse")
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
myeloma %>% select(molecular_group, treatment) # Select columns
myeloma %>% filter(molecular_group == "MMSET") # Select rows matching a condition
myeloma %>% filter(molecular_group %in% c("MMSET", "Cyclin D-1"))
myeloma %>% filter(molecular_group == "MMSET" & treatment == "TT2")
myeloma %>% filter(molecular_group == "MMSET" & TP53 < 1000)

MMSET_lowTP53 <- myeloma %>% filter(molecular_group == "MMSET" & TP53 < 1000)
```

You can also perform calculations

```

myeloma %>% summarise(mean(TP53)) # mean(myeloma$TP53)
myeloma %>% summarise(mean(TP53)/100) # divide by 100
myeloma %>% group_by(molecular_group) %>% summarise(mean(TP53))

# Create a new column with "mutate"
myeloma %>% mutate(TP53_expression = "low")

# Create a column based on a condition
# ifelse(expression, TRUE, FALSE)
myeloma <- myeloma %>% mutate(TP53_expression = ifelse(TP53 < 1500, "low", "high"))

```

Comparisons

Compare means

Two categories:

- T-test
- Wilcoxon test / Mann-Whitney U

Three or more categories:

- ANOVA
- Kruskal Wallis

```

# T-test
t.test(myeloma$TP53 ~ myeloma$event)

```

Welch Two Sample t-test

```

data: myeloma$TP53 by myeloma$event
t = 1.276, df = 115.08, p-value = 0.2045
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -72.4579 334.8488
sample estimates:
mean in group 0 mean in group 1
    1573.177      1441.981

```

```
# Normality test: Try doing it in each event subgroup
# shapiro.test(myeloma$TP53)

# ANOVA
aov(myeloma$TP53 ~ myeloma$molecular_group) %>% summary()
```

```

              Df      Sum Sq Mean Sq F value    Pr(>F)
myeloma$molecular_group    6  14251412 2375235    5.252 4.09e-05 ***
Residuals                 249 112613132  452262
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Non-parametric
# wilcox.test(myeloma$TP53 ~ myeloma$event)
# kruskal.test(myeloma$TP53 ~ myeloma$molecular_group)
```

Plot the comparison with [ggplot](#)

```
# ggplot(database, aes(x, y))+
# geom_boxplot --> you can change the geometry to geom_scatter, geom_bar, etc
# factor(event) will categorize the event variable
ggplot(myeloma, aes(x=factor(event), y=TP53))+
  geom_boxplot()
```

```
# Try to plot the molecular_group as y-axis
# Try to plot the distribution of TP53: use geom_histogram() and only x axis
```

Compare categories

- Chi-square
- Fisher exact test

```
# Event vs. molecular_group
# First check the proportions
table(myeloma$event) # single variable
```

```

  0    1
186  70
```

```
table(myeloma$event, myeloma$TP53_expression) # two categories
```

	high	low
0	89	97
1	27	43

```
prop.table(table(myeloma$event, myeloma$TP53_expression)) # in % or real proportions
```

	high	low
0	0.3476562	0.3789062
1	0.1054688	0.1679688

```
prop.table(table(myeloma$event, myeloma$TP53_expression), margin=1) # row sum
```

	high	low
0	0.4784946	0.5215054
1	0.3857143	0.6142857

```
prop.table(table(myeloma$event, myeloma$TP53_expression), margin=2) # column sum
```

	high	low
0	0.7672414	0.6928571
1	0.2327586	0.3071429

```
chisq.test(myeloma$event, myeloma$TP53_expression) # chi-square
```

Pearson's Chi-squared test with Yates' continuity correction

data: myeloma\$event and myeloma\$TP53_expression
X-squared = 1.4122, df = 1, p-value = 0.2347

Plot chi-square results

```
# Reproduce the following plots
ggplot(myeloma, aes(x=TP53_expression))+
  geom_bar()
```

```
ggplot(myeloma, aes(x=TP53_expression, fill = factor(event)))+
  geom_bar()
```

```
ggplot(myeloma, aes(x=TP53_expression, fill = factor(event)))+
  geom_bar(position = "fill")
```

Compare only numerical data

- Correlation
- Linear regression

```
# Correlation
cor.test(myeloma$TP53, myeloma$CCND1, method = "pearson") # or spearman
```

Pearson's product-moment correlation

```
data: myeloma$TP53 and myeloma$CCND1
t = -1.3651, df = 254, p-value = 0.1734
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.20579215  0.03765255
sample estimates:
      cor
-0.08534341
```

```
lm(myeloma$TP53 ~ myeloma$CCND1) # %>% summary()
```

Call:

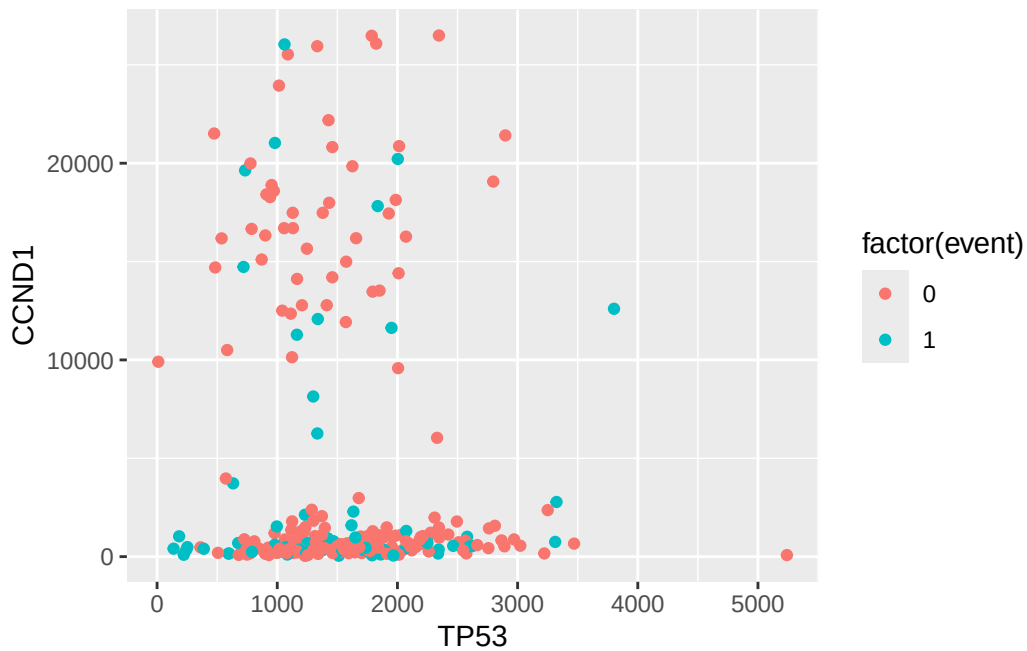
```
lm(formula = myeloma$TP53 ~ myeloma$CCND1)
```

Coefficients:

```
(Intercept) myeloma$CCND1
 1574.50847      -0.00828
```

Plot numerical data

```
ggplot(myeloma, aes(x=TP53, y=CCND1, color=factor(event)))+  
  geom_point()
```



This work is licensed under a
[Creative Commons Attribution-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-sa/4.0/).

